

Aniket Agarwal

✉ agarwalaniket1999@gmail.com in aniket-agarwal1999 aniket-agarwal1999 aniket-agarwal1999.github.io ☎ +1-878-834-9269

Education

Carnegie Mellon University - School of Computer Science

Master of Science in Robotics | **CGPA: 3.90/4.00**

Pittsburgh, PA

Aug. 2023 - July 2025

Indian Institute of Technology Roorkee

Integrated Master of Science in Applied Mathematics | **CGPA: 7.82/10**

Roorkee, India

July 2017 - May 2022

Experience

Gatik

Machine Learning Engineer | *Autonomy & Robotics*

Aug 2025 - Present

Mountain View, California

- Part of Perception team at Gatik AI. Developed an automated deterministic labeling pipeline to generate dense occupancy labels from time-synchronized RGB-LiDAR data, improving data efficiency and ground-truth quality for perception models.
- Leading efforts to build a scalable scene representation pipeline for generating real-time occupancy maps, enabling richer spatial reasoning for downstream planning tasks.

Carnegie Mellon University

Research Assistant | *Advisor: Prof Laszlo Jeni*

Nov 2023 - July 2025

Pittsburgh, Pennsylvania

- Built multimodal perception system for **DARPA Triage Challenge** integrating RGB, thermal, and audio streams for remote casualty assessment, achieving 4th place among 12 competing teams.
- Led the fine-tuning of LLM and TTS models for injury and verbal alertness assessment, quantized them for edge efficiency, and spearheaded development of real-time ROS inference pipelines deployed on NVIDIA Jetson systems.
- MSR Thesis Project:** Developed a multimodal human mesh recovery (HMR) framework fusing RGB, LiDAR, and IR data using a transformer-based Modality Unifier, enabling robust 3D pose estimation under degraded sensing. Achieved up to *14% lower MPJPE* than single-modality baselines and successfully deployed the model on Jetson Orin for real-world autonomy tasks & casualty motor-alertness detection. [Report Link](#)

Microsoft

Data & Applied Scientist | *MSAI Team*

July 2022 - Aug 2023

Hyderabad, India

- Built and deployed Learning-to-Rank models serving 100M+ daily queries for Outlook email/message ranking, improving relevance metrics (NDCG@10) by 10% and driving 5% increase in user engagement.
- Engineered feature importance-based dimensionality reduction for FastRank, reducing model latency by 20% while maintaining ranking quality, enabling real-time inference at scale.
- Migrated production LTR pipeline from Aether to Azure ML, implementing automated retraining workflows and validation pipelines that maintained NDCG parity while reducing deployment overhead by 60%.

Princeton University

Research Collaboration | *Advisor: Prof Karthik Narasimhan*

Oct. 2021 - Jun 2022

New Jersey, USA

- Developed automated benchmark generation pipeline for long-form video understanding, processing 10K+ hours of sports footage with temporal action detection and event segmentation using SOTA video transformers.
- Created and released novel LVU-Sports benchmark with comprehensive human evaluation, identifying 70% performance gaps in existing models and enabling new research directions in long-form video reasoning

Publications

- Aniket Agarwal**, Laszlo A Jeni. "Multimodal Human Mesh Recovery for Stand-off Triage in Mass Casualty Scenarios" *CMU MSR Thesis* — *In Submission at ICRA 2026* [[Project Page](#)]
- Aniket Agarwal***, Alex Zhang*, Karthik Narasimhan, Igor Gilitschenski, Vishvak Murahari, Yash Kant. "Building Scalable Video Understanding Benchmarks through Sports." *Data-Centric ML Research Workshop at International Conference on Learning Representations (DMLR@ICLR)*, 2024. [[Project Page](#)]
- Jun Chen, **Aniket Agarwal**, Sherif Abdelkarim, Deyao Zhu, Mohamed Elhoseiny. "RelTransformer: A Transformer-Based Long-Tail Visual Relationship Recognition." *Computer Vision and Pattern Recognition Conference (CVPR)*, 2022. [[Paper Link](#)]
- Aniket Agarwal***, Sherif Abdelkarim*, Panos Achlioptas, Jun Chen, Jiaji Huang, Boyang Li, Kenneth Church, Mohamed Elhoseiny. "Exploring Long Tail Visual Relationship Recognition with Large Vocabulary." *International Conference on Computer Vision (ICCV)*, 2021. [[Project Page](#)]

Technical Skills

Research Interests: Multi-modal perception, 3D Computer Vision, efficient neural representations, large-scale dataset curation, Vision-Language Models, Autonomous Systems

Frameworks/Libraries: PyTorch, PyTorch Lightning, PyTorch3D, Huggingface, PCL, TensorRT, ONNX, ROS2, OpenCV

Languages: Python, C++, CUDA, Bash